Bank Marketing Dataset: Analysis Report

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github.com/ziqing05/DS701-midterm

Q1: Ranking Clients Before a Call

Question: Can we rank later-period clients before dialing to capture more subscribers than random selection within the same contact budget?

Approach

We preserved the chronological order of 41,188 records: the earliest 60% for training, the next 20% for validation, and the final 20% for testing. We excluded `duration` and conservatively excluded `campaign`, leaving 18 inputs. Encoding and scaling were fitted on training data only.

Logistic regression provided a linear reference; random forests captured nonlinear relationships. Validation Lift@10% was 0.460 for logistic regression, **1.217 for a depth-5 forest**, and 1.118 for a depth-10 forest. We selected the depth-5 forest (200 trees, minimum leaf size 20), refitted on the earliest 80%, and tested on the final 20%. The same configuration was separately trained on a stratified random 80% split (seed 42) for comparison.

We selected the top 10% of clients by predicted score. **Lift@10%** is their subscription rate divided by the overall rate in the same test set; random selection has expected lift 1. **Recall@10%** is the proportion of all subscribers captured.

Results

Metric	Time-based (primary)	Random (comparison)
Test records / selected clients	8,238 / 824	8,238 / 824
Overall subscription rate	30.83%	11.26%
Subscribers captured	**487**	414
Random expected capture	254.06	92.82
Precision@10%	**59.10%**	50.24%
Recall@10%	**19.17%**	44.61%
Lift@10%	**1.917**	4.460

The chronological model captured about **233 more subscribers than random selection in expectation**, a **91.7% increase** at the same contact budget. Selecting 10% of clients captured 19.17% of subscribers. Random-split lift was higher, but its overall subscription rate was lower; differences in periods, prevalence, and training composition affect this comparison.

Limitations

Validation-half lifts were 1.076 and 1.396, suggesting time-varying improvement; these are development diagnostics, not independent confirmation. Earlier EDA accessed the full dataset, while this run selected the model before final testing. Without client IDs, performance on entirely unseen people remains unverified. Historical ranking gains do not establish additional subscriptions or profit.

Conclusion: Under a 10% contact budget, the model improved subscriber ranking in later-period records, with gains varying over time.

Q2: Predicting Subscription Before a Call

Question: Can we predict whether a new client will subscribe, using only information available before dialing?

Approach

We used Q1's 18 pre-call inputs and the same splits (chronological 60/20/20; stratified random 80/20, seed 42). Q2 needs a yes/no decision, so we judged the `yes` class by precision, recall and F1. Accuracy is supplementary: always predicting `no` scores 69–89% on our evaluation sets.

Candidates were L2-regularized logistic regression ($C = 0.01, 0.1, 1$) and random forests (depth 5 and 10, as in Q1), each fitted on training rows only. Instead of resampling, each candidate's threshold maximized validation F1 (ties to the higher threshold). Validation F1 was 0.220–0.221 for logistic regression, 0.217 for the depth-10 forest, and **0.224 for the depth-5 forest** at threshold 0.107. We selected it, fixed the threshold, refitted on the earliest 80% and on the random 80%, and tested each on its 20%.

Baselines were the majority class (always `no`, F1 0) and the one-feature rule `poutcome = success`. Because F1 rises with the base rate, we also report flagging everyone, whose F1 is $2r/(1+r)$ at base rate r.

Results

On the chronological test, the model identified 1,936 of 2,540 subscribers, with 2,271 false positives. Its F1 exceeded the one-feature rule's by 0.11–0.15 (paired bootstrap interval) and flagging everyone by 0.10. Its accuracy was below the majority class on both splits, because the threshold trades false positives for recall.

Test set, `yes` class	Time-based (primary)	Random (comparison)
Subscription rate	30.83%	11.26%
Predicted yes / subscribers	4,207 / 2,540	1,625 / 928
Precision	46.02%	36.80%
Recall	**76.22%**	64.44%
F1	**0.574**	0.468
95% bootstrap interval	0.559–0.588	0.442–0.494
One-feature rule F1	0.444	0.296
Flag-everyone F1	0.471	0.202
Accuracy (majority class)	65.10% (69.17%)	83.53% (88.74%)

Limitations

Validation evidence was weak. The selected forest's validation F1 (0.224) was barely above flagging everyone (0.199), and candidates differed by at most 0.007, so the model choice was not decisive. Validation-half F1 was 0.204 and 0.257, while the share predicted `yes` fell from 51% to 24%. The fixed threshold labelled 51% of time-test clients `yes`, reflecting the higher later subscription rate. A diagnostic best threshold on each test set, not used for selection, would have raised F1 only from 0.574 to 0.577 (time-based) but from 0.468 to 0.517 (random). Bootstrap intervals reflect test-record variation, not retraining. As in Q1, there are no client IDs and EDA accessed the full dataset; predictions do not show that calling causes subscriptions.

Conclusion: Pre-call attributes support a yes/no prediction that beat every baseline on later-period records (F1 0.574 vs 0.444–0.471), but its margin over flagging everyone was small during validation, and a good threshold depends on the period's subscription rate.

Q3: SVD and Clustering

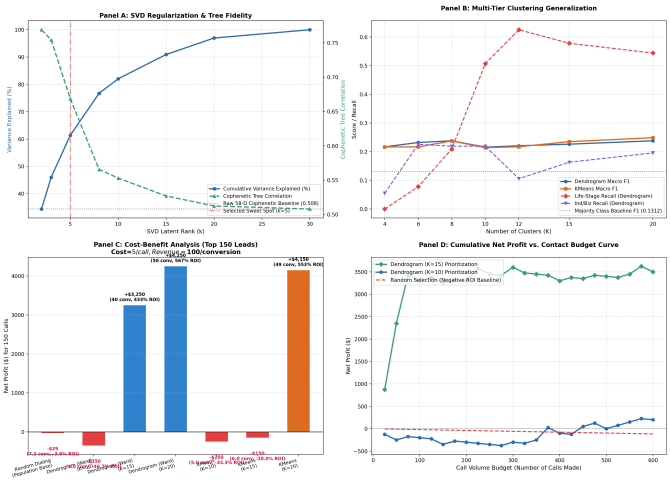
Question: Can we create logical, similar groups of prospective customers in order to better understand the customer base? In addition, can we assign high likelihood conversions based off of these groups without needing to run a predictor on each segment?

Approach

The goal here was to create logical groups of clusters which have some relation, in order to create groups which we can prioritize. Since there were many features, this was a two step process: first, an SVD breakdown to reduce the number of features, and then a dendrogram / KMeans evaluation to see what ROI looked like with some assumptions as to cost of call and reward from a success. This was then evaluated using cross validation. We used a 5-fold cross-validation approach, with the preprocessing and SVD fitted independently within each training fold to avoid information leakage between training and validation data. Additionally, we withheld duration and y which are not collected before a call. We used a sweep to evaluate both KMeans and Dendrogram, and found a $K = 20$ to be best for business performance for each, and an SVD latent rank of $k = 5$ for dimensionality reduction. This gives a 567% ROI compared to randomly sampling prospective customers. The CBA was calculated using a \$5 cost per call and a \$100 reward per success, which are estimates of the economics of the campaign. This allowed us to further tune for the best CBA, since cost can make F1 an incomplete metric.

The SVD step is particularly important because it reduces the original 58-dimensional feature space into five latent components. These components capture combinations of the original features rather than simply selecting five individual columns. We evaluated different latent ranks to find a balance between retaining information and preserving the underlying hierarchical distance structure. At $k = 5$, we retained approximately 61.4% of the original variance while achieving a cophenetic correlation of 0.669, compared to approximately 0.508 using the raw feature space. Importantly, the five latent dimensions do not limit us to five clusters: the subsequent hierarchical clustering can divide observations into 20 groups within this five-dimensional representation.

For the clustering evaluation, we tested different values of K (4, 6, 8, 10, 12, 15, and 20). Ward-linkage hierarchical clustering and KMeans were both evaluated. Since hierarchical clustering does not naturally provide a prediction method for unseen observations, we assigned validation observations to their nearest training-cluster centroid. KMeans used its standard prediction method. We then evaluated how well the resulting clusters generalized to different customer characteristics, including job tier, education, and housing status.



Evidence / Results

SVD of $k = 5$ reduced dimensionality from 58 to 5, which increased Cophenetic Fidelity from approximately 0.508

to 0.669. This was also included in a sweep which found it to be a reasonable balance between variance retention and tree fidelity. At this rank, approximately 61.4% of the original variance was retained. Higher ranks retained more variance, but progressively reduced cophenetic correlation toward the raw-data baseline. This suggests that dimensionality reduction helped remove noise and redundancy that could interfere with the hierarchical distance structure.

We ended up finding 4 distinct tiers of people, being: White Collar, Blue Collar/Services, Independent/Biz (including entrepreneurs and self-employed individuals), and "Life Stage" (Students/Retirees/Unemployed). The original job categories were consolidated into these broader groups to make the resulting segments easier to interpret. These tiers were then used to evaluate and explain the clustering results. Rather than treating the clusters as inherently meaningful labels, we examined which job tiers were disproportionately represented within each cluster relative to their overall population prevalence. This allowed us to associate clusters with the customer characteristics they represented.

The clustering results showed that $K = 20$ produced the strongest business outcomes for both methods. For the Dendrogram (Ward) approach, the top 150 prioritized calls produced 50 conversions, compared to approximately 7.2 expected conversions from random selection. This resulted in \$4,250 in net profit and a 566.7% ROI. KMeans with $K = 20$ produced 49 conversions and approximately 4,150 in net profit, with a 533% ROI. Both substantially outperformed random selection.

This demonstrates why optimizing purely for Macro F1 can be incomplete for this problem. A clustering configuration does not necessarily need to maximize overall classification performance to be economically useful. Instead, it needs to identify groups with sufficiently high conversion propensity to justify the cost of contacting them. Feeding our results into a CBA, we looked at calling high-propensity groups versus random dialing. This gave us an outcome of approximately +567% ROI for the Dendrogram approach, by prioritizing customers belonging to the highest-conversion-propensity clusters.

Overall, the results suggest that combining dimensionality reduction with hierarchical clustering can produce useful customer segments for targeted marketing. SVD $k = 5$ provides a more compact representation of the data, while Dendrogram $K = 20$ provides sufficiently granular segmentation to distinguish groups with substantially different conversion rates. The combination is therefore useful not only for describing customer populations, but also for making economically motivated decisions about which prospective customers should receive outreach.

Limitations

SVD and Clustering only has so much fidelity, especially in a case like this with a rare positive class. This means we will likely not get maximum ROI since we are grouping individuals into groups rather than evaluating them individually, so we cannot discriminate between many high interest individuals. Additionally, since this model uses SVD and is not temporally conscious, if the customer base drifts over time, the model will likely take a very long time to adjust. SVD also necessarily loses some information, which would not be lost if we used a less dimensionally dependent model like a random forest.